

HARDWARE DESIGN & VERIFICATION REPORT

Digital Bench Power Supply

Design Calculations, Circuit Simulations & Hardware Verification

Baran Güvendiren

Electronics Engineering Student

This document contains the complete theoretical calculations, simulation models, component selection justification and hardware verifications for the project.

Table of Contents

1	Calculations & Component Selection	2	1.2	Naming Convention & Notations	2
1.1	Overview	2	1.3	LM25117 - U1	2

1 Calculations & Component Selection

1.1 Overview

This section contains the theoretical calculations and engineering decisions used to determine the component values for each stage of the power supply.

The datasheet files containing manufacturer specifications and references used in these calculations are located in the `docs/datasheet/` directory of the project repository.

Note: Some calculations are based on idealized formulas with sufficient design margin to ensure system stability and flexibility in component selection.

1.2 Naming Convention & Notations

To maintain mathematical clarity across multiple stages, variables are named using the following conventions:

Table 1: Naming Convention

Notation Format	Description	Example
No Parentheses	Global board-level variable	V_{OUT} (Board output voltage)
With Parentheses	Component specific to a certain IC	$V_{OUT(U1)}$ (Output voltage specific to U1 - LM25117)

Note: The number inside the parentheses indicates which integrated circuit (IC) the variable belongs to.

Table 2: IC Reference Table

IC Number	Component Name	Description
1	LM25117	Main Buck Pre-Regulator
2	MIC29302	Main LDO
4	TPS54331	Auxiliary 5 V Buck Regulator
6	TPS54332	Auxiliary 12 V Buck Regulator

1.3 LM25117 - U1

1.3.1 Timer Resistor $R_{T(U1)}$

The switching frequency of the main buck regulator was set to 435 kHz via the $R_{T(U1)}$ pin resistor; this enables sufficient separation from the adapter's lower switching frequency (≈ 100 kHz) to minimize beat frequency noise, while preventing excessive switching losses at higher frequencies.

$$\begin{aligned}
 R_{T(U1)} &= \frac{5.20 \cdot 10^9}{f_{SW(U1)}} - 949 \\
 &= \frac{5.20 \cdot 10^9}{435 \cdot 10^3} - 949 \approx 11.01 \text{ k}\Omega
 \end{aligned} \tag{1}$$

Picked Value(s):

◇ $R_{T(U1)} = 11 \text{ k}\Omega$

1.3.2 Output Inductor $L_{O(U1)}$

The output inductance was calculated using a nominal mid-range voltage of $V_{OUT(TYP)} = 8\text{ V}$ to achieve a balanced ripple current trade-off over the full output voltage range. In addition, maximum inductor ripple ($I_{PP(U1)}$) current was selected as 20 % of the full load current.

$$\begin{aligned} L_{O(U1)} &= \frac{V_{OUT(TYP)}}{I_{PP(U1)} \cdot f_{SW(U1)}} \cdot \left(1 - \frac{V_{OUT(TYP)}}{V_{IN}}\right) \\ &= \frac{8.0}{0.55 \cdot 435 \cdot 10^3} \cdot \left(1 - \frac{8.0}{20.0}\right) \approx 20.06 \mu\text{H} \end{aligned} \quad (2)$$

Picked Value(s):

- ◇ $L_{O(U1)} = 20 \mu\text{H}$

1.3.3 Current Sense Resistor $R_{CS(U1)}$

Maximum output current capability ($I_{OUT(MAX)}$) was selected as 150 % of the full load current to account for tolerances.

Note: Since the LM25117 uses an emulated current ramp, it reads the valley current just before the high-side switch turns on. This means the controller is unaffected by the large leading-edge switching spikes. Therefore, a current sense filter was not used in this design to avoid propagation delay on current sensing.

$$\begin{aligned} R_{CS(U1)} &= \frac{V_{CS(TH)}}{I_{OUT(MAX)} + \frac{V_{OUT(TYP)} \cdot K}{f_{SW(U1)} \cdot L_{O(U1)}} - \frac{I_{PP(U1)}}{2}} \\ &= \frac{0.12}{4.13 + \frac{8 \cdot 1}{435 \cdot 10^3 \cdot 20 \cdot 10^{-6}} - \frac{550 \cdot 10^{-3}}{2}} \approx 25.16 \text{ m}\Omega \end{aligned} \quad (3)$$

$$\begin{aligned} P_{RCS(U1)} &= \left(1 - \frac{V_{OUT(MIN)(U1)}}{V_{IN}}\right) \cdot I_{OUT}^2 \cdot R_{CS(U1)} \\ &= \left(1 - \frac{1.25}{20}\right) \cdot 2.75^2 \cdot 25 \cdot 10^{-3} \approx 177.59 \text{ mW} \end{aligned} \quad (4)$$

Picked Value(s):

- ◇ $R_{CS(U1)} = 25 \text{ m}\Omega$
- ◇ $P_{RCS(U1)} = [\text{TBD}]$

1.3.4 Ramp Resistor and Capacitor $R_{RAMP(U1)}$ and $C_{RAMP(U1)}$

Current sense amplifier gain (A_{RAMP}) and K factor are both defined in the datasheet. K factor was set to 1 to prevent sub-harmonic oscillations and ensure one-cycle damping, as recommended in the datasheet.

Note: The value of $C_{RAMP(U1)}$ was set to the standard capacitor value of 1 nF.

$$\begin{aligned} R_{RAMP(U1)} &= \frac{L_{O(U1)}}{K \cdot C_{RAMP(U1)} \cdot R_{CS(U1)} \cdot A_{RAMP}} \\ &= \frac{20 \cdot 10^{-6}}{1 \cdot 1 \cdot 10^{-9} \cdot 25 \cdot 10^{-3} \cdot 10} = 80 \text{ k}\Omega \end{aligned} \quad (5)$$

Picked Value(s):

- ◇ $R_{\text{RAMP}(U1)} = 80 \text{ k}\Omega$
- ◇ $C_{\text{RAMP}(U1)} = 1 \text{ nF}$

1.3.5 Output Shunt Resistor R_S

The output current was measured by converting the load current ($I_{\text{OUT}(U1)}$) into an analog voltage signal via a dedicated shunt resistor and a current sense amplifier. The resulting analog signal was then digitized by the MCU's ADC. Since the LDO's input current is approximately equal to its output current, the shunt resistor was placed before the feedback node at the LDO input to prevent the voltage drop across the shunt resistor from affecting the regulated output voltage.

As low power consumption was not a primary concern for this project, the amplifier gain (A_S) was set to $50\times$ to minimize the amplification of measurement errors. In addition, the reference voltage ($V_{S(\text{REF})}$) was limited to 3 V, providing 300 mV headroom below the MCU supply voltage to reduce the impact of possible control errors.

Note: $R_{CS(U1)}$ was used to measure the inductor valley current for LM25117's emulated current-mode control, while R_S was used to calculate the output current for monitoring and setting CC-mode current limit.

$$\begin{aligned}
 R_S &= \frac{V_{S(\text{REF})}/A_S}{I_{\text{OUT}}} \\
 &= \frac{3.0/50}{2.75} \approx 21.82 \text{ m}\Omega
 \end{aligned} \tag{6}$$

$$\begin{aligned}
 P_{RS} &= I_{\text{OUT}}^2 \cdot R_S \\
 &= 2.75^2 \cdot 22 \cdot 10^{-3} \approx 166.38 \text{ mW}
 \end{aligned} \tag{7}$$

Picked Value(s):

- ◇ $R_S = 22 \text{ m}\Omega$
- ◇ $P_{RS} = [\text{TBD}]$